



UTM
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**Faculty of
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Section : 01

Project : Lab Assignment 1

Task : Daily Activities Recorded App

Lecturer : Dr. AHMAD NAJMI BIN AMERHAIDER NUAR

Prepared by:

No.	Name	Matric Number
1	Lee Yin Shen	A23CS0236

Daily Activities Recorder App

1.0 Objective

The main objective of this assignment is to build a working mobile and desktop application using the Flutter framework. The assignment required us to include several specific technical parts in the app, which include at least two text input fields (**TextField**), at least one button (**Button**), a scrolling list to show items (**ListView**), a way to update the screen when data changes (**setState**), and a feature to delete items. Beyond just meeting the basic requirements, the objective was also to create an app that is actually useful and looks professional. To do this, we aimed to add extra features, like making sure the data saves permanently even when the app is closed, and using pop-up calendars and clocks to make it very easy for the user to enter their information.

2.0 System Description

The Daily Activities Recorder is a digital diary and task manager. It is built using the Dart programming language and the Flutter toolkit. The system allows users to easily add new tasks, look at all their planned activities, change the details of existing tasks, and delete tasks they no longer need.

A very important part of this system is how it remembers information. Normally, when you close a simple app, all the data disappears. To fix this, the system uses a special tool called **shared_preferences** along with a data format called JSON. This means every time a user adds or changes a task, the app secretly saves that information directly into the phone or computer's memory. When the user opens the app again later, the system quickly loads all the saved tasks so nothing is ever lost.

3.0 Explanation of the Interface

The app is designed to be very simple and friendly to use. The visual interface is broken down into a few main areas:

3.1 The Main Dashboard

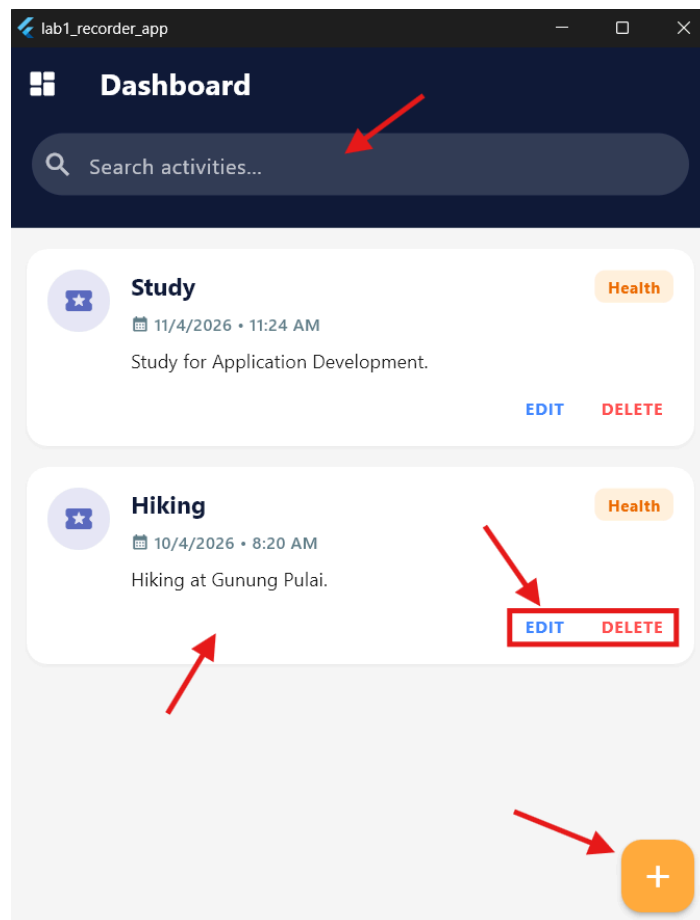


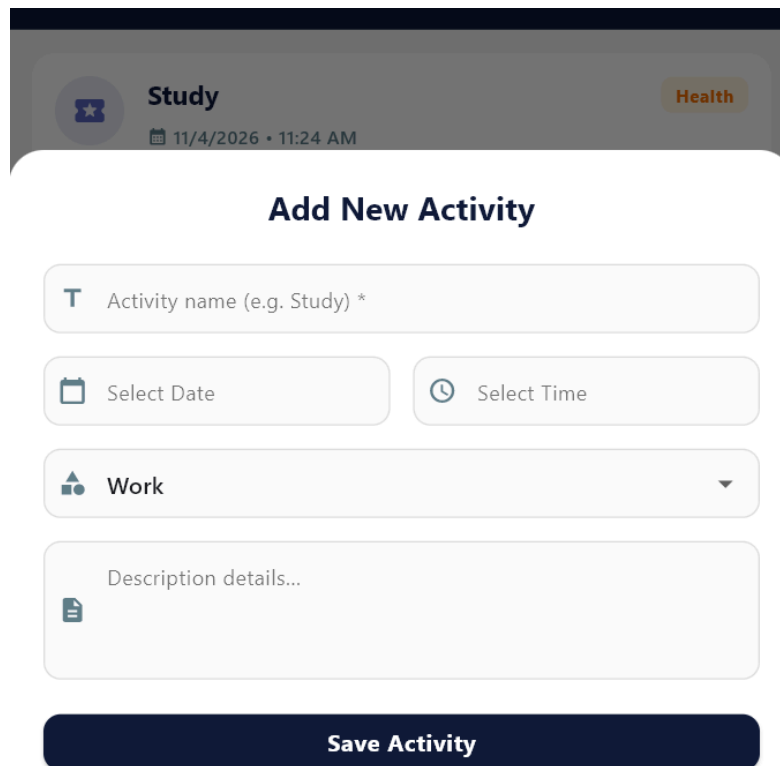
Figure 3.1: Main Dashboard

When you open the app, this is what you see.

- **Search Bar:** Right at the top, there is a search box. If you have many tasks, you can type a word here to quickly find a specific activity without scrolling.
- **Activity List:** The middle of the screen is a long, scrolling list (**ListView**). Every task you add appears here inside a neat, rounded box called a "card". Each card shows the name of the task, a small picture (icon) showing what type of task it is, the date, the time, and any extra notes you wrote.

- **Edit and Delete Buttons:** On the right side of every task card, there are two small buttons. One is a pencil icon to change the task, and the other is a trash can icon to delete it.
- **Add Button:** At the bottom right corner, there is a large, round "+" button. This is the main button used to trigger the "Add function" and make a new task.

3.2 Add Activity Function



The screenshot shows a mobile application interface for adding a new activity. At the top, there's a header bar with a star icon, the word 'Study', and a 'Health' tag. Below this, the title 'Add New Activity' is centered. The form consists of several input fields: a text field for 'Activity name (e.g. Study) *', two buttons for 'Select Date' and 'Select Time', a dropdown menu currently showing 'Work', and a larger text area for 'Description details...'. At the bottom of the form is a prominent dark blue button labeled 'Save Activity'.

Figure 3.2: Data Entry Form

When you click the “+” button, this is what you see.

- **Data Entry Form:** When the user clicks the Add button, a new form slides up from the bottom of the screen. This form uses multiple text input fields (using Flutter's **TextField** widget) to collect information. The user can type the name of their activity and a detailed description directly into normal input fields. Once finished, clicking the "Save" button uses the **setState** command to update the system and show the new activity on the Home Page.

3.3 Interactive Date & Time Pop Up, Category Drop Down List

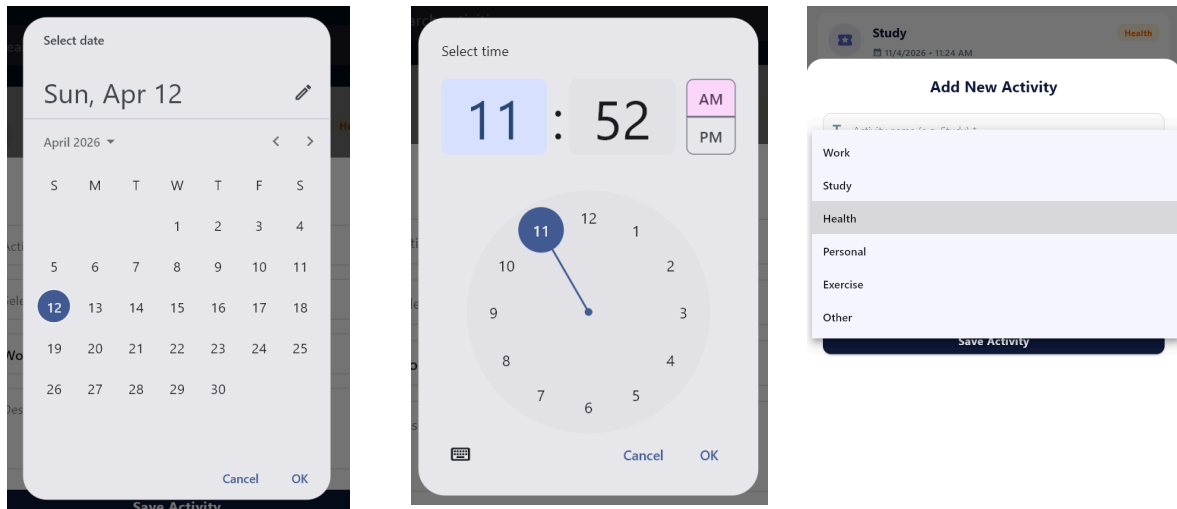


Figure 3.3: Interactive Date & Time Pop Up, Category Drop Down List

When you add activity detail, this is what you see.

- **Interactive Date & Time Pop Up:** To prevent typing mistakes, instead of manually typing "April 12" or "5:00 PM" into a blank box, tapping on these fields opens a beautiful, interactive pop-up calendar or clock. The user simply taps the exact day and time they want.
- **Category Drop Down List:** Instead of typing a category, the user clicks a box that drops down a list of pre-set choices (like Work, Study, or Health). If none of those fit, selecting "Other" smartly reveals an extra hidden text input field so they can type their own custom choice.

4.0 Screenshots of the app

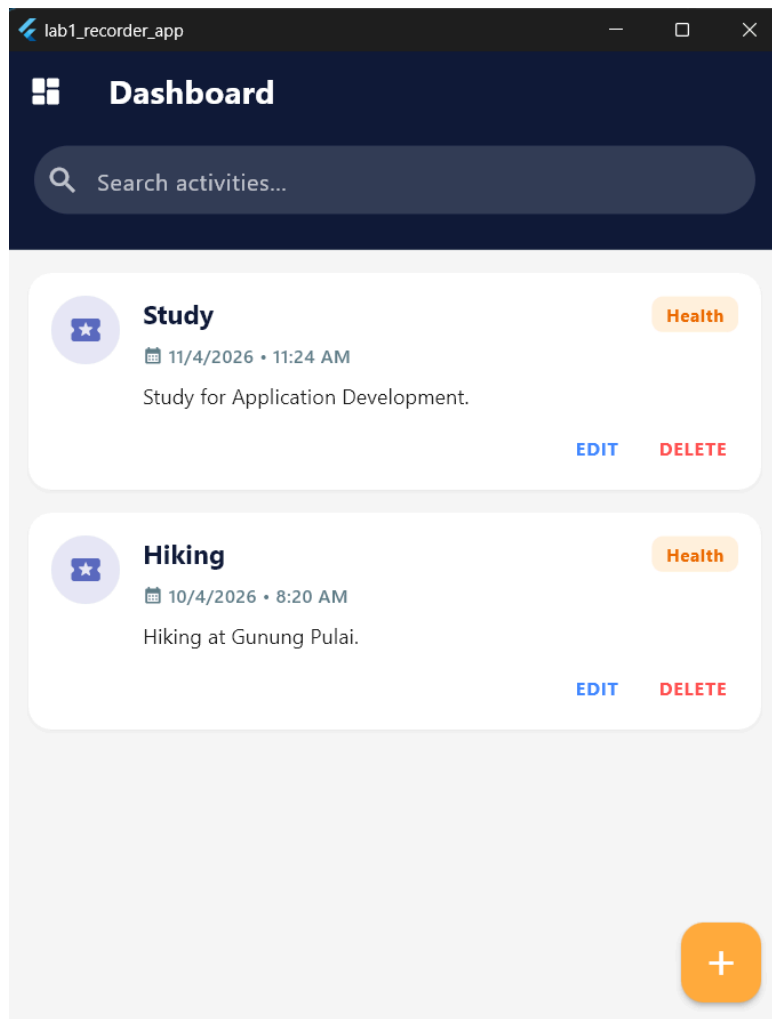
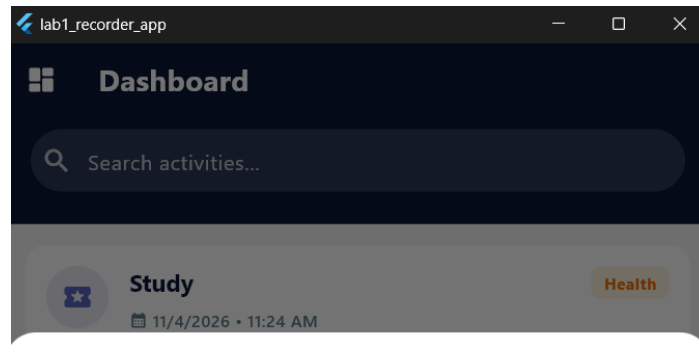


Figure 4.1: Home Page

The Home Page shows the search bar, the scrolling ListView of activities, the edit/delete buttons, and the floating Add button.



Add New Activity

Figure 4.2: Data Entry Form

The Add function shows the slide-up form with its multiple text input fields.

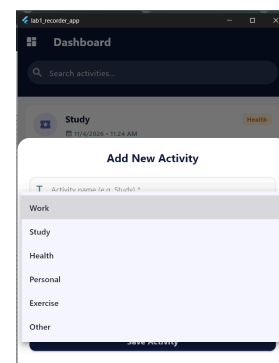
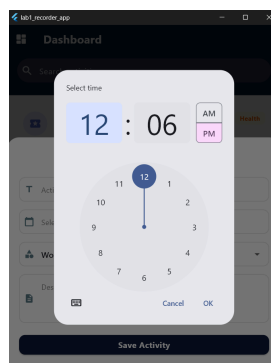
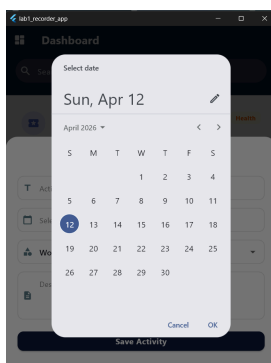


Figure 4.3, 4.4, 4.5: Date, Time, Category Input Field

The interactive pop-up calendar and clock that appears when entering the Date and Time input field. The Category Drop Down list showing the different preset options like Work and Study.

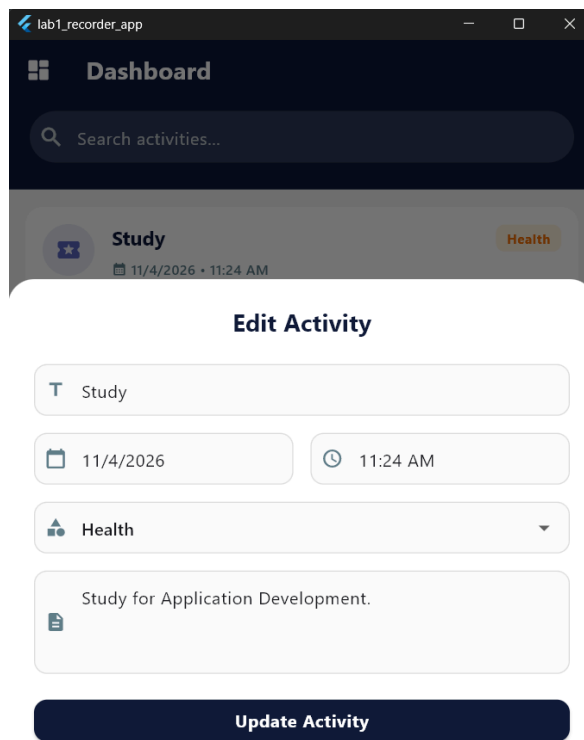


Figure 4.6: Editing Activity Detail

The edit button allows the user to edit and update the activity detail.

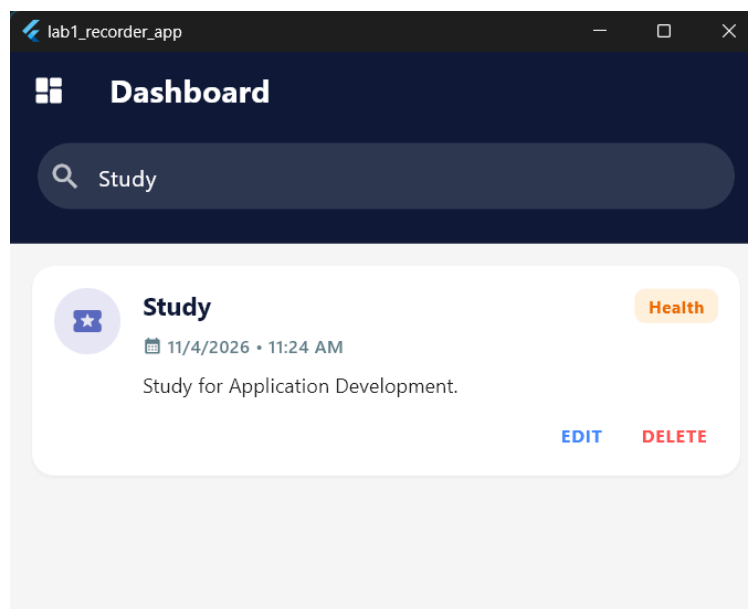


Figure 4.7: Search Bar

The search bar allows the user to search specific activities.

5.0 Conclusion

In conclusion, the Daily Activities Recorder successfully demonstrates how to build a fully functional and user-friendly mobile application. The core of the app is built on a StatefulWidget, which efficiently holds and manages the changing task data. By combining text input fields to gather data, interactive buttons to submit actions, and a delete function to remove unwanted items, the system functions seamlessly. Whenever a user interacts with the app, the setState command correctly triggers the screen to update and refresh the ListView so the user always sees the most current information. Additionally, by adding permanent data saving, real-time search filtering, and smart interactive menus like the dropdown lists and date pickers, this project shows a strong understanding of how to build digital tools that are reliable, highly functional, and extremely practical for everyday use.